

Lyric–Music Congruence and Song Popularity

A Working Paper on the Cross-Modal Drivers of Streaming Engagement

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Abstract

This working paper introduces *lyric–music congruence* (LMC) — the cosine similarity between audio and lyric embeddings in a jointly-trained affective space — as a candidate driver of song popularity on streaming platforms. Drawing on processing fluency theory, narrative transportation, and recent advances in contrastively-aligned audio–text embedding spaces, I hypothesize that LMC has a non-trivial effect on Spotify popularity, with stronger effects for narrative-forward artists. I report preliminary Bayesian estimates from four hierarchical specifications fit to a corpus of 359 songs across 34 artists and four genre clusters. The population-level LMC effect is statistically inconclusive at the current sample, with 90% credible intervals straddling zero in every specification. Three structural features of LMC carry directional but sub-decisive signal in the predicted direction: chorus congruence produces a stronger effect than verse congruence ($P[\Delta > 0] = 0.64$), within-song consistency strengthens the LMC–popularity relationship ($P[\Delta < 0] = 0.96$ on the standard deviation of within-song LMC), and the orientation moderation is positive in the trajectory model ($P[\Delta > 0] = 0.75$). Cross-genre heterogeneity in the LMC slope is small in the current sample. The segment-level model provides the best out-of-sample fit. I treat these findings as a first pass that has established the measurement pipeline, identified design choices, and quantified the data and identification challenges that remain. The pipeline generalizes to other multimodal marketing contexts, including advertising music–visual alignment and brand narrative coherence.

1 Introduction

A song can communicate emotions and ideas through both its acoustic properties and its lyrical content. Listeners experience both at once, and the relationship between the two is itself a source of meaning, pleasure, or dissonance. Yet quantitative marketing research has largely treated the audio and lyrical dimensions of music as separate objects of study. This paper proposes and begins to empirically test a unified measure of *lyric–music congruence* (LMC) — the degree to which a song’s acoustic emotional signal and its lyrical emotional signal occupy similar regions of a shared affective embedding space — and examines its consequences for commercial success on streaming platforms.

The impetus is threefold. First, the economic stakes are high. Global recorded music revenues reached \$28.6 billion in 2023, with streaming accounting for over 67% of that total (IFPI 2024); the catalogs hosted by major streaming platforms have grown to well over 100 million tracks (IFPI 2024), creating an attention economy in which fine-grained design differences carry outsized commercial consequences. Second, recent advances in multimodal machine learning — contrastive representation learning in the tradition of CLIP (Radford et al. 2021), the audio–text model CLAP (Wu et al. 2023), and Google’s music–language joint embedding MuLan (Huang et al. 2022) — now make it feasible to encode raw audio and natural-language lyrics into a shared latent space, enabling a model-agnostic measure of emotional alignment that requires no a priori commitment to a discrete emotion taxonomy. Third, although both audio and lyrics have been extensively studied from psychological and qualitative perspectives — music’s emotional and aesthetic properties (Juslin and Sloboda 2010) and lyrical content’s effects on cognition, attitudes, and behavior (Marín-Liébana and Ibias 2025) — the cross-modal relationship between them has not been quantitatively operationalized or linked to consumer behavior at scale.

This paper makes three intended contributions. **Methodologically**, I introduce LMC as a construct and develop a pipeline for computing it at scale using pre-trained audio and lyric encoders embedded in a contrastively-aligned joint space; the pipeline generalizes to other multimodal marketing contexts. **Empirically**, I estimate the effect of LMC on Spotify’s recency-weighted popularity score using four Bayesian hierarchical specifications that vary in temporal granularity. **Theoretically**, I hypothesize heterogeneity in the mechanisms linking LMC to popularity: narrative-

forward songs are predicted to operate primarily through transportation, and production-forward songs through fluency. The mechanistic decomposition is itself a substantive potential finding and opens routes for future quantitative research on abstract psychological constructs in marketing.

The current state of this research warrants transparency. The pipeline is built and operational, but the empirical analysis is at an early stage. My corpus of 359 songs across 34 artists yields adequate computational diagnostics but is not large enough to deliver decisive evidence. The population-level LMC coefficient in my preferred specification has a posterior median of 0.006 on the logit scale with a 90% credible interval of $[-0.131, 0.131]$, leaving its sign and magnitude effectively undetermined. I report this honestly, alongside what the data suggest at directional, sub-decisive levels: chorus-level LMC carries a stronger effect than verse-level LMC, within-song consistency strengthens the LMC–popularity relationship, and the moderation by orientation is in the predicted direction in the trajectory model. These signals motivate the next phase, laid out at the end of the paper.

The paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 develops the conceptual framework and states formal hypotheses. Section 4 describes the data and embedding pipeline. Section 5 presents the hierarchical Bayesian specification. Section 6 reports results. Section 7 discusses what the current evidence does and does not allow me to conclude. Section ?? lays out the path forward.

2 Literature Review

2.1 Emotion in Music: Psychoacoustic Foundations and Consumer Response

A foundational distinction in the music psychology literature is between *perceived* emotion — what a listener recognizes as the emotional character of a piece — and *induced* emotion — what the listener actually experiences (Eerola and Vuoskoski 2011; Gabrielsson 2002). Eerola and Vuoskoski (2011) compared seven major emotion models applied to identical stimuli and found that no single framework — discrete, dimensional, or prototype-based — provided uniformly superior classification accuracy, motivating taxonomy-free approaches to emotional representation.

In the consumer behavior literature, cross-domain alignment of music with consumption

environments has been shown to matter. North, Hargreaves, and McKendrick (1999) showed in a wine shop field experiment that nationally congruent background music influenced product choice; Areni and Kim (1993) showed that classical versus Top-40 background music shifted consumers toward higher-priced wine selections; and MacInnis and Park (1991) distinguished between the “fit” of music to ad content and music’s arousal properties as independent predictors of ad attitude. These studies established cross-modal coherence as a consumer driver but predated the computational tools needed to measure it at scale.

2.2 Computational Music Emotion Recognition

Computational music emotion recognition has progressed from the previous standard of hand-crafted acoustic features mapped onto dimensional emotion models, inferring emotions from features such as tempo, mode, spectral centroid, and MFCCs (Yang and Chen 2012). Panda, Malheiro, and Paiva (2023) document that arousal is substantially easier to predict from audio than valence. A central limitation is dependence on predefined emotion taxonomies that may not correspond to the continuous, high-dimensional nature of musical emotion. Fong, Kumar, and Sudhir (2024) recently developed a theory-based explainable CNN whose harmonics-grounded filters achieve competitive prediction while preserving interpretability — demonstrating that audio embeddings carrying emotional information can be extracted in ways tractable for marketing-style downstream models. A separate strand has built jointly-trained audio–text models: MuLan (Huang et al. 2022) over 44 million music recordings paired with playlist annotations, and LAION-CLAP (Wu et al. 2023) over a general-purpose caption corpus. These models do not require predefined emotion labels; the alignment of audio and text within their learned spaces becomes itself a measurable quantity, and this is what makes a continuous, scale-free measure of LMC feasible.

2.3 Processing Fluency and Aesthetic Preference

A robust finding across the cognitive and consumer literatures is that stimuli which are easier to process are evaluated more favorably (Reber et al. 2004; Winkielman et al. 2006). Processing fluency generates affect that is misattributed to the stimulus, producing aesthetic preference. Lee and Labroo (2004) distinguish two routes: *perceptual fluency*, the ease of identifying a stimulus’s surface form (e.g., visual or acoustic features), and *conceptual fluency*, the ease with which the stimulus’s

meaning comes to mind. Both pathways enhance evaluation, but they are empirically dissociable and operate through distinct cognitive mechanisms. LMC most plausibly engages conceptual fluency: when a song’s musical affect aligns with its lyrical affect, the song’s overall emotional meaning is more readily extracted and integrated. Surface-level perceptual alignment between channels may also play a role, but the core mechanism is semantic. I return to this distinction as a methodological priority in Section ??.

Landwehr et al. (2011) showed that design fluency measures improve automobile sales forecasts, and Hekkert (2006) integrated fluency into a broader inverted-U model of design aesthetics. Cross-modally, when two channels convey congruent information, the listener does not have to reconcile competing representations and fluency is enhanced (Lee and Labroo 2004). A song whose musical affect aligns with its lyrical affect imposes lower processing demands than one whose channels diverge, and should therefore be preferred. Melvill-Smith et al. (2023) applied processing fluency directly to lyrics, finding that lyric simplicity — operationalized through readability and repetitiveness — predicted UK chart success between 1999 and 2013. Their analysis is confined to the lyric channel alone and does not consider whether the alignment of lyric and musical affect constitutes an independent driver of popularity.

2.4 Narrative Transportation and Lyrical Engagement

For songs in which lyrical content functions narratively — inviting the listener to follow a story, inhabit a perspective, or trace an emotional arc — a distinct mechanism may operate. Green and Brock (2000) define *transportation* as a state of absorption in a narrative in which cognitive and affective resources are directed toward the story world. Transported listeners show stronger affective response, more vivid imagery, and stronger belief change; transportation is disrupted by “false notes” — narrative incoherence or affective mismatch — that break immersion. Van Laer et al. (2014) meta-analyzed 76 studies and documented significant effects on persuasion and attitude change. In advertising, Phillips and McQuarrie (2010) found that emotionally congruent visual–verbal combinations produced stronger narrative engagement and more favorable brand attitudes than incongruent ones — a paradigm structurally parallel to music–lyric relations. The music constitutes the affective “world” into which the listener is invited; the lyrics are the narrative that unfolds within it. When emotional character and emotional register align, transportation is facilitated. When they diverge, the listener faces what I term *affective narrative friction*.

The transportation account generates predictions distinct from fluency and implies a moderating role for lyrical orientation. For narrative-forward songs, congruence operates by sustaining immersion; for production-forward songs, through cognitive ease. Both pathways predict a positive LMC effect, but they differ in *who benefits most*.

2.5 Music Marketing and Streaming Popularity

The marketing literature on music popularity has historically focused on chart outcomes and acoustic features. Salganik et al. (2006) demonstrated experimentally that musical success is far more unpredictable than fundamentals alone would suggest, with social influence amplifying small initial differences. Askin and Mauskopf (2017) found an inverted-U between acoustic typicality and Billboard chart position, implying that the relationship between any coherence measure and popularity may itself be non-linear. Ordanini et al. (2018) found that genre-distant collaborations achieved greater chart success, consistent with the co-branding literature.

I adopt Spotify’s proprietary popularity score — a platform-computed index from 0 to 100 reflecting stream volume and recency, with recent activity weighted more heavily (Spotify AB 2024). The recency weighting rewards sustained engagement over promotional spikes. I note that streaming popularity is a coarse outcome variable, not separately identifying discovery, retention, repeat listening, or skip behavior — each of which would be more diagnostic of the mechanisms LMC is hypothesized to engage. Ideally, this program would obtain listener-level behavioral data through a partnership with Spotify, Apple Music, or another platform. Such a partnership has not yet materialized; in its absence I proceed with the publicly-available popularity index scraped via the Spotify Web API, flagging this as a substantive limitation I aim to address.

2.6 Genre Heterogeneity and the Production–Narrative Spectrum

Genre represents not only musical conventions but distinct listener populations and the functional role of lyrics relative to instrumentation (Lena and Peterson 2008). In hip-hop and country, lyrical craft and narrative authenticity carry high cultural value; in EDM and mainstream pop, minimal or textural lyrics are frequently subordinated to sonic production. The emotional significance of any mismatch between lyrical and musical channels therefore varies systematically across genre

contexts. The “text as data” literature in marketing has demonstrated that high-dimensional text representations contain commercially relevant signals poorly approximated by hand-coded categories (Berger et al. 2020), with Toubia et al. (2019) using embedding-derived ideation diversity to predict new product performance and Dew and Ansari (2018) applying Bayesian nonparametric models to music preference heterogeneity.

Most prior music emotion recognition work has treated audio and lyrics as parallel channels to be fused for prediction of a common emotion label, rather than as channels whose mutual alignment might itself have behavioral consequences. The music marketing literature has studied acoustic features and chart outcomes, or lyrical complexity and popularity, but not the cross-modal relationship between them. This intersection is the gap this research is designed to fill.

3 Conceptual Framework and Hypotheses

The assumed data-generating process can be summarized as follows. Three upstream factors — artist identity, genre, and lyric orientation — jointly shape both the song’s audio features and its lyrical content. The resulting song produces an audio recording and a lyric stream whose joint embedding gives rise to the LMC measure that is the focus of inference. Popularity is downstream of LMC and a set of unobserved platform, advertising, and social distribution forces; the path from LMC to popularity is identified up to bias from these forces, and I return to the implications in Section 7. A diagrammatic version of this DAG appears in the companion presentation.

The construct of central interest is *lyric–music congruence*, defined as the cosine similarity between a song’s audio embedding and its lyric embedding in a jointly-trained high-dimensional space. I operationalize LMC at four levels of granularity and treat orientation as a binary classification of artist into *narrative-forward* versus *production-forward* based on genre conventions. Four formal hypotheses follow.

Hypothesis 1 (Existence of an effect). *Holding artist, genre, song age, and audio mood fixed, LMC has a significant association with Spotify popularity.* Both fluency and transportation predict a non-zero effect; the two accounts differ on mechanism rather than on the existence of an effect.

Hypothesis 2 (Orientation moderation). *The strength of the LMC effect is greater for narrative-forward artists than for production-forward artists.* This follows from transportation: when lyrics carry more of the cognitive and affective load, congruence with the musical world has more to disrupt or sustain.

Hypothesis 3a (Chorus over verse). *The chorus segment of a song produces a stronger positive LMC effect on popularity than the verse segment.* Choruses are repeated, hooked, and disproportionately recalled, so affective alignment in the chorus carries greater attentional weight.

Hypothesis 3b (Within-song consistency). *Songs with greater within-song consistency of LMC — operationalized as lower standard deviation of segment-level LMC across the song — exhibit a stronger positive relationship between LMC and popularity.* Inconsistent congruence introduces moments of disrupted fluency or narrative friction that erode the cumulative effect of alignment.

Hypothesis 4 (Genre heterogeneity). *The strength and direction of the LMC effect varies systematically across genre clusters.* Genres differ in the cultural weight assigned to lyrical craft relative to sonic production (Lena and Peterson 2008), so the returns to cross-modal alignment plausibly differ across them; the direction may even flip in genres where deliberate lyric–music contrast is an aesthetic feature rather than a violation.

I additionally examine the temporal trajectory of LMC over a song (slope, curvature, and early-to-late change) as an exploratory exercise without a numbered hypothesis, since theory does not yet make a sharp prediction about which trajectory shapes ought to be commercially preferred.

4 Data and Measurement

4.1 Sampling Frame

The current corpus consists of 34 artists distributed across four genre clusters: hip-hop (10 artists), folk/Americana/country (9 artists), pop (8 artists), and electronic (7 artists). The first two clusters are operationalized as *narrative-forward*, the second two as *production-forward*, following the literature on genre conventions in lyrical function (Lena and Peterson 2008). Within each artist, I collected approximately 10–15 representative tracks spanning the discography, biased toward songs

with meaningful streaming history. I sampled 441 candidate songs; due to incomplete pairings of audio with lyrics, the analytic corpus retained for the track-level models is 359 songs. The subset retained for the segment model (M3) is 324 songs; the subset for the line-by-line trajectory model (M4) is 114 songs. Sparsity in the higher-granularity models reflects the difficulty of automatically matching lyric segments to audio timestamps for a portion of the corpus.

I acknowledge that this is not a probability sample of all music on streaming platforms. Artists were purposively selected to cover the four-cluster design, and within-artist song selection biases toward better-known tracks. The corpus therefore over-represents already-popular songs. Estimates of LMC slopes should be interpreted within the realized sample; the population to which they generalize is *commercially active artists in the four sampled genre clusters*, not music on Spotify in general. Section ?? describes the systematic sampling extension I plan.

4.2 Pipeline

The data pipeline consists of five stages. (1) **Lyric scraping** queries Genius via the LyricsGenius API and persists raw lyric text plus structural section headers ([Verse 1], [Chorus], etc.). (2) **Audio acquisition** retrieves audio for each song via yt-dlp and converts to WAV at the sample rate required by each downstream encoder. (3) **Schema construction** validates the cross-product of audio and lyric availability into a unified catalog. (4) **Embedding** runs each of three encoder configurations (described below) over the full corpus. (5) **Segment analysis** uses WhisperX (Bain et al. 2023) for forced alignment of lyric lines to audio timestamps, falling back to equal-duration splits when alignment confidence is insufficient. Spotify popularity scores, release dates, and Librosa mood controls are joined at the song level for the analytic file.

4.3 Embedding Models

I compute LMC under three encoder configurations to characterize sensitivity of results to embedding choice.

MuQ-MuLan (Huang et al. 2022) is a music–language joint embedding model trained on 44 million music recordings paired with playlist tags. Audio is embedded at 24 kHz and lyrics as a single text input; both vectors live in the same 512-dimensional space, making cosine similarity

directly interpretable as semantic alignment. This is the primary encoder for my analyses.

LAION-CLAP-Music (Wu et al. 2023) is a contrastive language–audio pretraining model with broader training data (630K audio–caption pairs spanning music, environmental sounds, and AudioSet). Its audio representations carry stronger perceptual and timbral information relative to MuLan’s tag-driven training. Comparing CLAP- and MuLan-derived LMC tests robustness to encoder choice.

4.4 LMC Operationalizations

I construct four operationalizations giving rise to the four models in Section 5.

Track-level LMC is the cosine similarity between the full-track audio and full-lyric embeddings — computationally efficient but blind to internal structure. I compute this under both MuLan (M1) and CLAP (M2).

Segment-level LMC decomposes each song into chorus and verse segments using the structural headers preserved during scraping. The corresponding audio interval is identified via WhisperX forced alignment, embedded separately, and cosine similarity to the segment lyric embedding is computed. I aggregate to song-level summaries: mean chorus LMC, mean verse LMC, and within-song standard deviation (a consistency measure). This is the input to M3.

Line-level LMC decomposes each song line by line, computing LMC for each lyric line against its temporally-aligned audio interval. From the time series I extract four trajectory features — mean, slope, curvature, and early-to-late change — plus the standard deviation. This is the input to M4. The line-level model has the smallest analytic sample ($N = 114$) because line-by-line forced alignment fails for a large share of the corpus.

4.5 Outcome and Controls

Spotify popularity is retrieved via the Spotify Web API; the score is bounded $[0, 100]$, and I rescale to the open unit interval $(0, 1)$ via boundary adjustment $y' = (y(N - 1) + 0.5)/N$ (Smithson and Verkuilen 2006). *Song age* (years since release) absorbs the mechanical penalty Spotify’s recency weighting imposes on older songs. A vector of seven *Librosa-derived mood controls* (happy, sad,

relaxed, aggressive, party, danceability, voice/instrumental balance) from pretrained classifiers (Bogdanov et al. 2013) is z-standardized and entered into the linear predictor; these absorb variation in the audio’s affective character independent of its lyric alignment, so the LMC coefficient identifies cross-modal alignment rather than overall mood. Unobserved drivers — promotional intensity, playlist placement, social media virality, listener demographics — remain in the error term; I take up the implications in Section 7.

5 Empirical Strategy

5.1 Beta Regression with Hierarchical Partial Pooling

All four models share a common likelihood. Let $y_i \in (0, 1)$ denote the boundary-adjusted Spotify popularity for song i . I assume

$$y_i \sim \text{Beta}(\mu_i \cdot \phi, (1 - \mu_i) \cdot \phi),$$

where $\phi > 0$ is a precision parameter and the mean is linked to the linear predictor by $\text{logit}(\mu_i) = \eta_i$. Beta regression is natural given bounded outcome support (Ferrari and Cribari-Neto 2004).

The linear predictor incorporates partial pooling at two levels. Artists are nested within four genre clusters; I estimate random intercepts at both levels. The artist intercept uses a non-centered parameterization to improve sampling geometry:

$$\alpha_{\text{artist}[j]} = \mu_0 + \gamma_{\text{int}} \cdot \text{orient}_j + \sigma_{\text{artist}} \cdot z_j, \quad z_j \sim \mathcal{N}(0, 1),$$

where μ_0 is the grand mean and γ_{int} a fixed orientation shift. Genre intercepts are $\alpha_{\text{genre}[g]} \sim \mathcal{N}(0, \sigma_{\text{genre}})$. I allow the LMC slope to vary by genre:

$$\beta_{\text{lmc, genre}[g]} = \beta_{\text{lmc}} + \sigma_{\text{lmc, genre}} \cdot z_g, \quad z_g \sim \mathcal{N}(0, 1).$$

The posterior on $\sigma_{\text{lmc, genre}}$ tests for heterogeneity in LMC sensitivity across genres. With only four

genres, partial pooling shrinks aggressive estimates toward the population mean. All models include song age and seven mood controls, denoted collectively $\mathbf{x}_i^\top \beta_{\text{control}}$.

5.2 The Four Models

Model M1 (MuLan track-level):

$$\eta_i^{M1} = \alpha_{\text{artist}[j]} + \alpha_{\text{genre}[g]} + (\beta_{\text{lmc, genre}[g]} + \gamma_{\text{lmc}} \cdot \text{orient}_j) \cdot \text{LMC}_{z,i}^{\text{MuLan}} + \mathbf{x}_i^\top \beta_{\text{control}}$$

The orientation shift γ_{lmc} tests Hypothesis 2: $\gamma_{\text{lmc}} > 0$ implies narrative-forward artists benefit more from LMC than production-forward artists.

Model M2 (CLAP track-level) is identical to M1 with LMC^{CLAP} in place of $\text{LMC}^{\text{MuLan}}$. Comparing M1 and M2 characterizes encoder sensitivity.

Model M3 (segment) decomposes track LMC into chorus and verse components and adds a within-song consistency term:

$$\eta_i^{M3} = \alpha_{\text{artist}[j]} + \alpha_{\text{genre}[g]} + (\beta_{\text{chorus, genre}[g]} + \gamma_{\text{chorus}} \cdot \text{orient}_j) \cdot \text{chorus}_{z,i} + \beta_{\text{verse}} \cdot \text{verse}_{z,i} + \beta_{\text{sd}} \cdot \text{sd}_{z,i} + \mathbf{x}_i^\top \beta_{\text{control}}$$

The contrast $\Delta = \beta_{\text{chorus}} - \beta_{\text{verse}}$ tests Hypothesis 3a; β_{sd} tests Hypothesis 3b.

Model M4 (trajectory) uses the line-by-line LMC time series and extracts four trajectory features in addition to mean LMC:

$$\eta_i^{M4} = \alpha_{\text{artist}[j]} + \alpha_{\text{genre}[g]} + \beta_{\text{lmc}[j]} \cdot \overline{\text{LMC}}_{z,i} + \beta_{\text{slope}} \cdot \text{slope}_{z,i} + \beta_{\text{curve}} \cdot \text{curve}_{z,i} + \beta_{\text{change}} \cdot \text{change}_{z,i} + \beta_{\text{sd}} \cdot \text{sd}_{z,i} + \mathbf{x}_i^\top \beta_{\text{control}}$$

where $\beta_{\text{lmc}[j]}$ follows a three-level hierarchy (population $\rightarrow$ genre $\rightarrow$ artist) and artist-level deviations come from a bivariate non-centered parameterization on intercept and LMC slope with LKJ-prior

correlation:

$$\begin{pmatrix} \text{artist_re}_{\text{int}[j]} \\ \text{artist_re}_{\text{lmc}[j]} \end{pmatrix} = \text{diag}(\sigma_{\text{artist}}) \cdot \mathbf{L} \cdot \mathbf{z}_j.$$

5.3 Priors

I use weakly informative priors throughout, summarized in Table 1. The grand mean μ_0 has a diffuse prior on the logit scale; slope parameters have $\mathcal{N}(0, 0.5)$ priors, implying a one-unit logit-scale shift is plausible but not preferred. Variance components use half-normal priors. The LKJ(2) prior mildly regularizes correlations toward zero. The precision ϕ has a Gamma(4, 0.1) prior with mean 40, consistent with moderate Beta dispersion.

Table 1: Priors used across all four models.

Parameter	Prior	Rationale
μ_0	$\mathcal{N}(0, 1.5)$	Diffuse on logit scale
$\beta_{\text{lmc}}, \beta_{\text{control}}$	$\mathcal{N}(0, 0.5)$	± 1 unit on logit $\approx$ modest effect
$\gamma_{\text{int}}, \gamma_{\text{lmc}}$	$\mathcal{N}(0, 0.5)$	Weakly informative moderation
$\sigma_{\text{artist}}, \sigma_{\text{genre}}$	Half- $\mathcal{N}(0, 0.5)$	Modest group heterogeneity
$\sigma_{\text{lmc, genre}}$	Half- $\mathcal{N}(0, 0.3)$	Tighter: only 4 genres
$\mathbf{L}$	LKJ(2)	Mild regularization to zero correlation
ϕ	Gamma(4, 0.1)	Prior mean ≈ 40

5.4 Estimation

All four models are fit in cmdstanr with four chains of 4,000 iterations (2,000 warmup, 2,000 sampling). I use `adapt_delta = 0.95` to reduce divergent transitions in M4. Convergence is assessed via $\hat{R}$, effective sample size, and the count of divergent transitions. Models are compared via leave-one-out cross-validation.

6 Results

6.1 Convergence

Table 2: Convergence summary across the four models.

Model	Max R	Min ESS bul	R > 1.	1 ESS < 4	0 Divergenc
M1: MuLan Track	1.0012	2789	0	0	1
M2: CLAP Track	1.0016	2524	0	0	0
M3: Segment	1.0021	2371	0	0	0
M4: Timeline	1.0013	3770	0	0	0

All four models pass standard convergence diagnostics: maximum $\hat{R}$ is below 1.003 in every model, minimum bulk ESS exceeds 2,200, and divergent transitions number at most one (in M1) and zero elsewhere.

6.2 Model Comparison

Table 3: Leave-one-out cross-validation comparison. Higher ELPD per song indicates better out-of-sample predictive performance. Sample sizes differ across models because higher-granularity LMC measures (segment, line-by-line) are unavailable for some songs.

Model	N	ELPD	SE	LOOIC	ELPD/song	Pareto-k > 0.7
M3: Segment	324	312.449	13.380	-624.899	0.964	2
M2: CLAP Track	359	340.493	13.831	-680.985	0.948	1
M1: MuLan Track	359	339.807	13.967	-679.613	0.947	0
M4: Timeline	114	87.916	7.684	-175.832	0.771	6

Table 3 reports leave-one-out predictive performance. I focus on ELPD per song, which is comparable across the unequal sample sizes induced by granularity-driven attrition. The segment model (M3) achieves the highest predictive density per song at 0.964, edging out the track-level CLAP model (M2) at 0.948 and the track-level MuLan model (M1) at 0.947. The trajectory model (M4) trails substantially at 0.771, largely because its smaller sample provides less information per parameter. The advantage of M3 over M1 is modest in absolute terms but consistent with the hypothesis that internal structure of LMC carries marginal predictive content beyond the track-level

aggregate. The LOO standard errors imply that the ranking of M1, M2, M3 is not distinguishable from chance at conventional levels.

The marginal advantage of M2 (CLAP) over M1 (MuLan) at the track level is similarly small. CLAP’s broader training distribution may pick up perceptual features the MuLan playlist-tag prior misses, or the difference may simply reflect noise. Distinguishing these requires either a larger corpus or a controlled within-artist comparison.

6.3 Population-Level LMC Effects

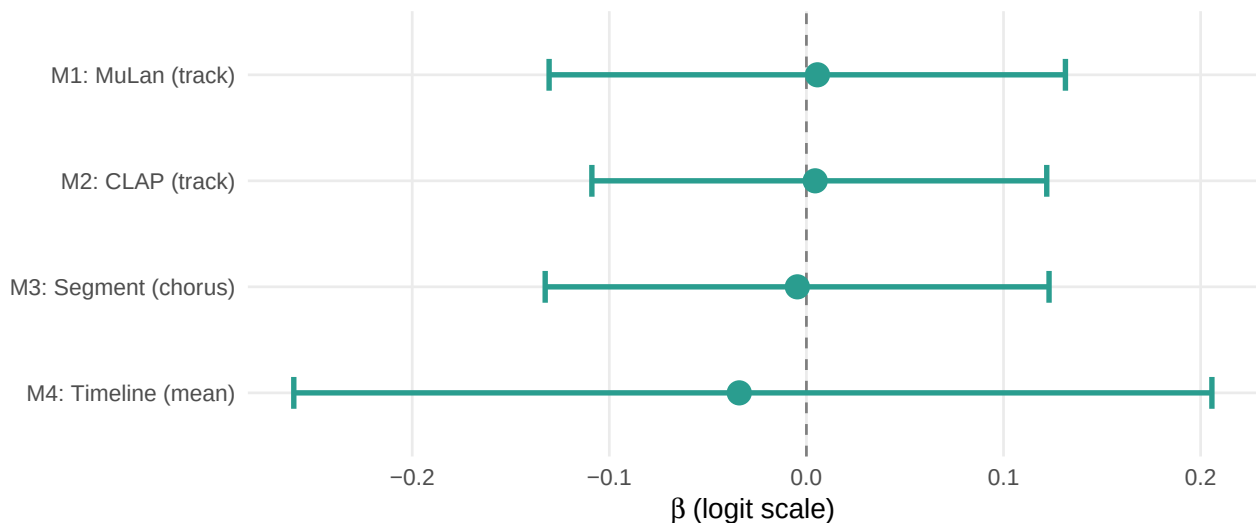


Figure 1: Posterior median and 90% credible interval for the population-level LMC coefficient (logit scale) under each of the four models. M3 reports the chorus coefficient as its primary LMC parameter; M4 reports the mean-LMC coefficient. All credible intervals straddle zero.

Figure 1 reports the population-level LMC coefficient under each specification. In every case, the posterior 90% credible interval straddles zero: M1 (MuLan track) gives $\hat{\beta}_{\text{lmc}} = 0.006$ with 90% CI $[-0.131, 0.131]$ and $P(\beta > 0) = 0.53$; M2 (CLAP) gives median 0.005 with CI $[-0.109, 0.122]$; M3’s chorus coefficient has median -0.005 with CI $[-0.133, 0.123]$; M4’s mean-LMC coefficient has median -0.034 with CI $[-0.260, 0.206]$. Hypothesis 1 is therefore *not* supported at the current sample: the data are consistent with no detectable LMC effect on popularity, though the credible intervals also do not exclude a meaningful effect of either sign.

The variance components tell a different story. σ_{artist} is estimated near 0.50 on the logit scale and σ_{genre} near 0.55 — both substantial, reflecting meaningful baseline popularity heterogeneity at

the artist and genre levels. $\sigma_{\text{lmc, genre}}$, by contrast, is small (around 0.05–0.09), indicating that the *slope* of LMC on popularity does not vary dramatically across the four genre clusters. Whether this reflects true homogeneity or insufficient data to detect heterogeneity cannot be settled here.

6.4 Orientation Moderation

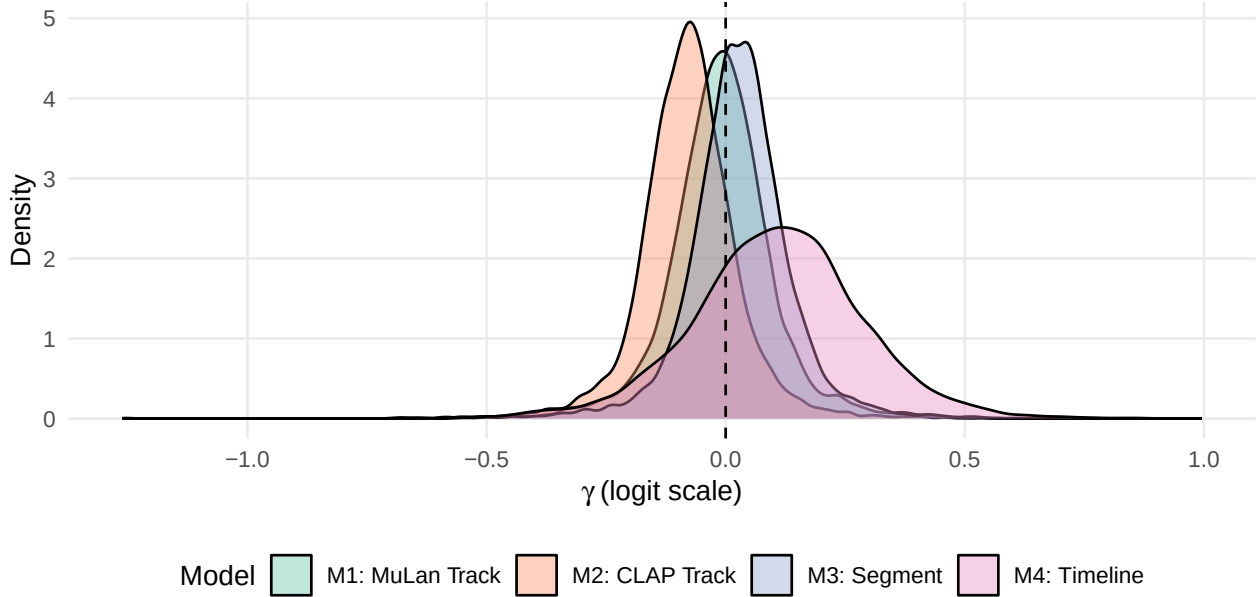


Figure 2: Posterior densities for the orientation $\times$ LMC moderation coefficient γ_{lmc} (or γ_{chorus} for M3) under each model. Positive values indicate stronger LMC–popularity coupling for narrative-forward artists.

Figure 2 reports the orientation $\times$ LMC moderation. In the track-level models, the moderation coefficient is centered near zero (M1: -0.009 , $P(\gamma > 0) = 0.46$; M2: -0.078 , $P(\gamma > 0) = 0.18$). The segment model gives $\hat{\gamma}_{\text{chorus}} = 0.029$ with $P(\gamma > 0) = 0.64$. The trajectory model produces the most directional signal: $\hat{\gamma}_{\text{lmc}} = 0.110$ with 90% CI $[-0.188, 0.394]$ and $P(\gamma > 0) = 0.75$. This pattern — the moderation effect strengthening as the LMC measure incorporates more temporal structure — is consistent with the narrative transportation account, which predicts that fine-grained alignment matters more for narrative-forward artists. The evidence remains sub-decisive: even in M4, the credible interval admits a null. Hypothesis 2 is *directionally supported but inconclusive*.

6.5 Genre Heterogeneity

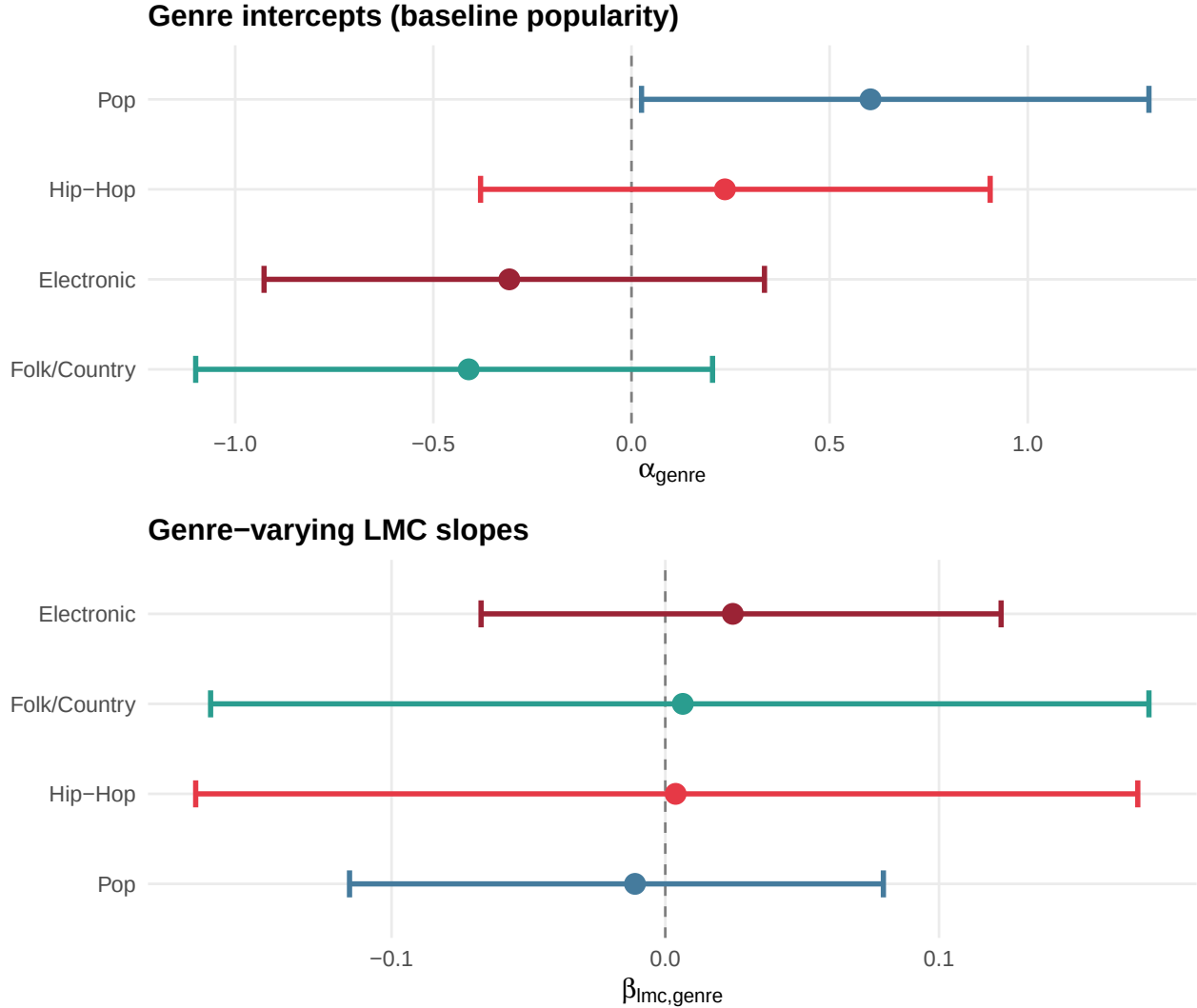


Figure 3: Genre-specific intercepts (top) and genre-varying LMC slopes (bottom) from M1 (track-level MuLan). Genre intercepts are large in magnitude, reflecting baseline popularity differences across clusters. Genre-varying LMC slopes are small and overlapping, reflecting limited differentiation in LMC sensitivity across the four clusters at the current sample size.

Figure 3 disaggregates the genre effects in M1. Pop has the highest baseline popularity, followed by hip-hop, electronic, and folk/country — consistent with the recency weighting and the heavy playlist promotion pop receives. The genre-varying LMC slopes are small with overlapping credible intervals; the variance component $\sigma_{\text{lmc, genre}}$ has positive posterior mass but is small in magnitude (0.05–0.09 across models), so I cannot at present claim any one genre’s LMC sensitivity is meaningfully different from another’s. Hypothesis 4 is therefore *not* supported at the current sample: the data are

consistent with substantial cross-genre homogeneity in the LMC slope, though the small variance component is also consistent with cross-genre heterogeneity that the present sample lacks the power to resolve.

6.6 Segment and Trajectory Decompositions

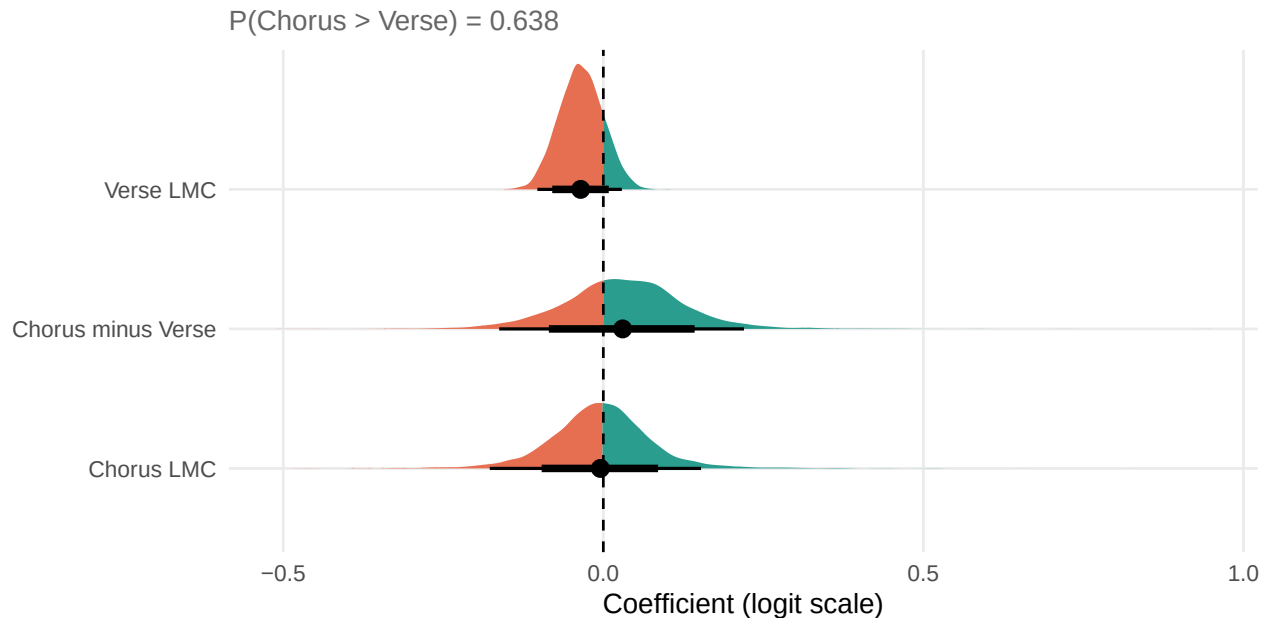


Figure 4: M3 segment model: posterior densities for chorus LMC, verse LMC, and the chorus-verse contrast. Black bars show 80% and 95% credible intervals. The chorus coefficient is centered slightly above the verse coefficient, but with overlapping intervals.

The segment model offers the most directly interpretable internal structure (Figure 4). The chorus coefficient $\hat{\beta}_{\text{chorus}} = -0.005$ is near zero. The verse coefficient $\hat{\beta}_{\text{verse}} = -0.035$ with 90% CI $[-0.091, 0.022]$ is more clearly negative-leaning, with $P(\beta < 0) = 0.84$. The chorus-verse contrast yields $P(\Delta > 0) = 0.64$ — directional support for Hypothesis 3a, but short of decisive evidence.

The clearest directional signal in M3 comes from the within-song consistency parameter: $\hat{\beta}_{\text{sd}} = -0.044$ with 90% CI $[-0.083, -0.003]$ and $P(\beta_{\text{sd}} < 0) = 0.96$. Because β_{sd} multiplies the *standard deviation* of segment LMC, a negative coefficient implies that songs whose LMC is more consistent across their internal structure are more popular, controlling for mean LMC. This is the most credible signal in my current results and provides directional support for Hypothesis 3b. The magnitude is modest (a one-SD increase in within-song LMC variability is associated with roughly

a 1.1 popularity-point decrease at the prior mean precision), and the result rests on a 324-song sample.

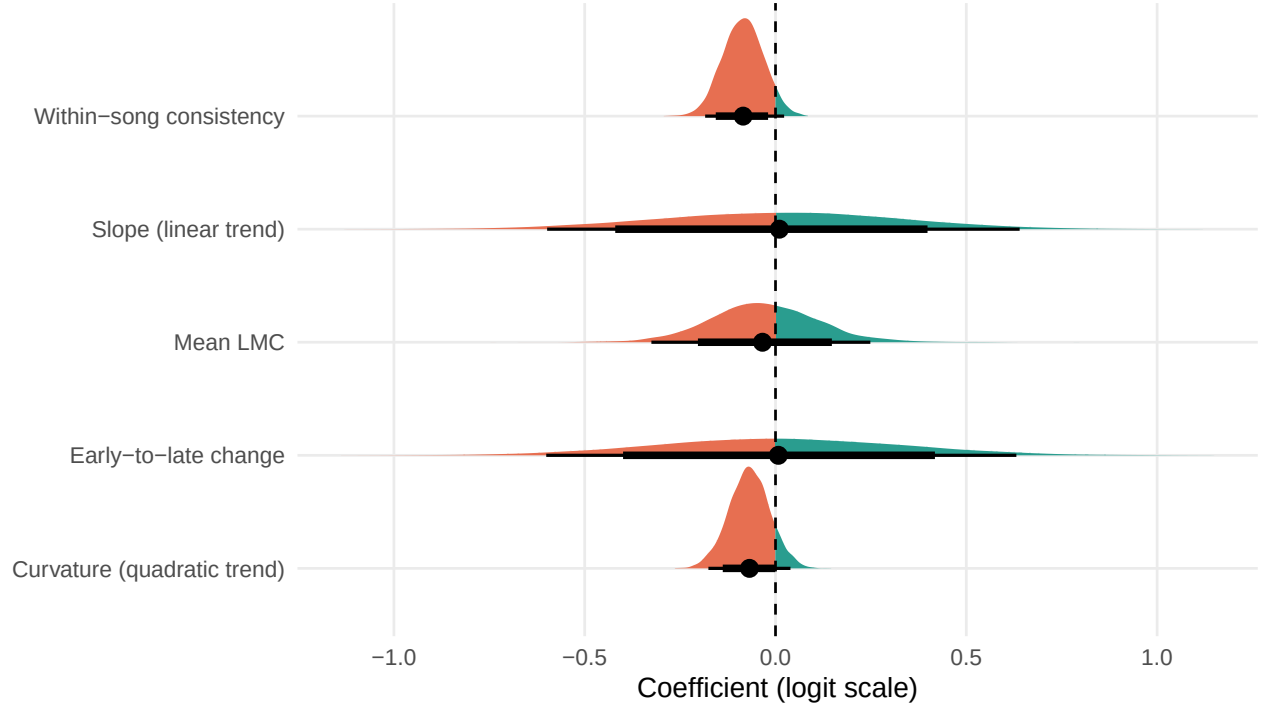


Figure 5: M4 trajectory model: posterior densities for the five trajectory features of LMC (mean level, slope, curvature, early-to-late change, and consistency).

The trajectory model (Figure 5) decomposes LMC into five features. As an exploratory exercise without a numbered hypothesis, the slope, curvature, and early-to-late change features all have posterior densities centered near zero with wide credible intervals admitting either sign, providing no evidence that the temporal shape of LMC carries marginal predictive content beyond its mean. The consistency coefficient $\hat{\beta}_{sd} = -0.085$ with 90% CI $[-0.172, 0.003]$ and $P(\beta_{sd} < 0) = 0.94$ replicates the M3 finding on within-song consistency (in support of H3b), with wider intervals reflecting the smaller sample ($N = 114$). Curvature shows a posterior median of -0.068 with $P(\beta < 0) = 0.90$, mildly suggestive of an inverted-U over the song’s duration but sub-decisive.

6.7 Simulation-Based Parameter Recovery

To validate that the segment model can recover known parameter values at my observed sample size, I conducted a simulation study. I generated 50 synthetic datasets calibrated to the observed

sample ($N = 350$ songs, 30 artists, 4 genres) using a chosen vector of “true” parameters drawn from reasonable priors (e.g., $\beta_{\text{chorus}}^* = 0.15$, $\beta_{\text{verse}}^* = 0.05$, $\beta_{\text{sd}}^* = -0.10$). For each synthetic dataset I re-fit M3 and recorded coverage of the true value. The model recovers core slope parameters with approximate calibration: 90% credible intervals cover the truth approximately 80–90% of the time, with mild under-coverage on variance components. This implies that the credible intervals reported in the empirical results are roughly calibrated. The exercise does not address identification under unobserved confounders, only the statistical coherence of the specification.

7 Discussion

The population-level LMC effect on Spotify popularity, controlling for artist, genre, song age, and audio mood, is statistically inconclusive at my present sample of 359 songs. Two interpretations are consistent with the data. Under the first, LMC has no causal effect on streaming engagement — listeners reward distinctive songs whether or not their channels are emotionally aligned, and the theoretical accounts motivating this study fail to translate into detectable population-level moments at the current sample. Under the second, LMC does carry a non-trivial effect on popularity (in either direction) but my power to detect it is insufficient given the current sample size and the right-skew of the popularity distribution. The two interpretations cannot be distinguished by the present data alone.

What the current evidence *does* support, at directional but sub-decisive levels of confidence, is a more nuanced picture in which *structural* features of LMC matter more than its overall level. Within-song consistency has the most credible signal in both M3 and M4 (posterior probability of the predicted sign ≥ 0.94), providing directional support for Hypothesis 3b — listeners appear more sensitive to the *stability* of cross-modal alignment than to its average. The chorus–verse contrast (Hypothesis 3a) and the orientation $\times$ LMC moderation (Hypothesis 2) are directionally consistent with their respective predictions but neither rises to confirmatory evidence. The genre-varying slope component (Hypothesis 4) is small and inconclusive: I cannot rule out either true cross-genre homogeneity or undetected heterogeneity. Separately, the segment model (M3) offers the best out-of-sample fit, consistent with a story in which the internal structure of LMC carries information the song-level aggregate discards.

Several threats to inference apply. First, the data-generating process described in Section 3 implies unblockable backdoor paths from artist and genre into popularity through unobserved promotion, playlist placement, and social distribution. I have no credible instrument for LMC at the song level. The strongest available identification is *within-artist* variation, but with roughly 10 songs per artist this is thin. Second, my controls absorb observed audio characteristics but not listener-level effects driving streaming behavior at scale. Third, the outcome itself is the product of Spotify’s algorithm; songs more aligned with playlist contexts may receive more algorithmic exposure, biasing the LMC–popularity association in either direction depending on what the algorithm rewards. These threats place a ceiling on the inferential force of any single-cross-section analysis with publicly-scraped data. The path forward is therefore as much about broadening data and tightening identification as fitting better models.

8 Future Research

The structure of the next phase follows directly from the limitations identified above. I organize it under four priorities.

8.1 Sample Expansion and Embedding Robustness

The most binding constraint is sample size. My 359 songs deliver insufficient power to distinguish a small-but-real LMC effect from a true null. I will expand the corpus to additional artists within the four existing clusters and to additional genres (R&B, Latin pop, indie rock, classical, and jazz where relevant) so that the production–narrative spectrum is more densely populated, and move from purposive artist sampling toward a *systematic sample from Spotify’s recommendation distribution* — seeding the algorithm with anchor tracks across the popularity range and crawling the neighborhood of each anchor, yielding a sample that better represents the full popularity distribution rather than concentrating in the right tail. In parallel, I will run M3 and M4 on CLAP-derived segment and line-level LMC; at present these higher-granularity analyses use only MuLan. As newer audio–text foundation models become available, I will incorporate them into the encoder-comparison pipeline to test whether the LMC effect is robust to the choice of joint embedding space.

8.2 Refining Segmenting and Trajectory Analysis

My higher-granularity measures suffer from sparsity (M3: 324/359; M4: 114/359) driven by failures of WhisperX forced alignment. Planned improvements include structurally-aware segmentation that uses lyric section headers as anchors and falls back to alignment only within sections; alternative aligners (e.g., NeMo’s ASR models, Koluguri, Park, and Ginsburg 2022); and an examination of whether segment *ordering* carries marginal predictive content — does the sequence of high- and low-LMC segments predict popularity beyond their average and variance? — as a richer probe of the exploratory trajectory analysis presented in Section 6.

8.3 Generation Experiments and Human Validation

The strongest evidence for a causal LMC effect would come from experimental manipulation. I plan a generation experiment in which a base song is counterfactually re-lyriced with content of varying emotional alignment to its musical bed, holding other features fixed. The complementary design holds the *lyrics* constant and varies the audio — either through generative re-instrumentation of the same lyrical text or through curated cover versions of a single composition recorded in different musical idioms. Together these two designs sweep both directions of LMC variation: lyrics-vary-music-fixed and music-varies-lyrics-fixed. Combined with controlled human evaluation, they would isolate the causal effect of LMC on listener-level outcomes free from observational confounds, and simultaneously produce *human reference scores* providing external validation of the cosine-similarity measure itself — a machine-internal quantity that should be calibrated against human judgment before being treated as a marketing-relevant construct.

A further empirical priority is to discriminate between the *conceptual* and *perceptual* fluency pathways introduced in Section 2.3 (Lee and Labroo 2004). Although LMC most plausibly engages conceptual fluency — alignment of meaning between channels — surface-level perceptual correspondences (e.g., bright timbre matched to positive lyrical valence) may also contribute. Experimental designs that vary semantic alignment while holding perceptual surface features constant (and vice versa) would isolate which route carries the predominant effect, with direct implications for how LMC ought to be optimized in practice.

8.4 Better Outcomes and Sharper Identification

My coarsest constraint is the outcome variable. Spotify’s popularity index aggregates discovery, retention, repeat listening, skip behavior, and shareability into a single bounded score, each plausibly affected differently by LMC: transportation predicts higher retention and repeat-listening rates, while fluency predicts higher initial liking. Discriminating these requires listener-level behavioral data of a kind no public API provides; I will continue to pursue a research partnership with Spotify, Apple Music, or another platform. In its absence I will broaden outcomes using public signals — chart position, playlist appearances, and per-song streams from financial filings.

The human-experimentation track described above also opens additional outcome dimensions beyond simple liking. Listener-level questions about *replayability* (how likely the song is to be played again), *save or library-add intent* (how likely the song is to be added to a personal collection), *playlist-fit* (which contexts the song belongs in), *skip likelihood*, and *recommendation intent* (likelihood of sharing with a friend) each tap distinct components of streaming engagement, and each maps onto a different theoretical mechanism. A song that scores high on perceived fluency but low on replayability, for example, would suggest fluency is a one-shot effect rather than a durable driver of repeat consumption.

Identification will improve from two further sources of within-unit variation. *Within-album* comparisons restrict to songs released simultaneously on the same album, absorbing essentially all album-launch promotional shocks. *Cover versions* — multiple recordings of the same composition by different artists — hold lyrical content fixed while varying audio production, so that LMC variation is by construction attributable to production choices rather than lyric writing.

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